

Communication Complexity: Communication Lower Bounds For Multi-Party Graph Traversal

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Motivation

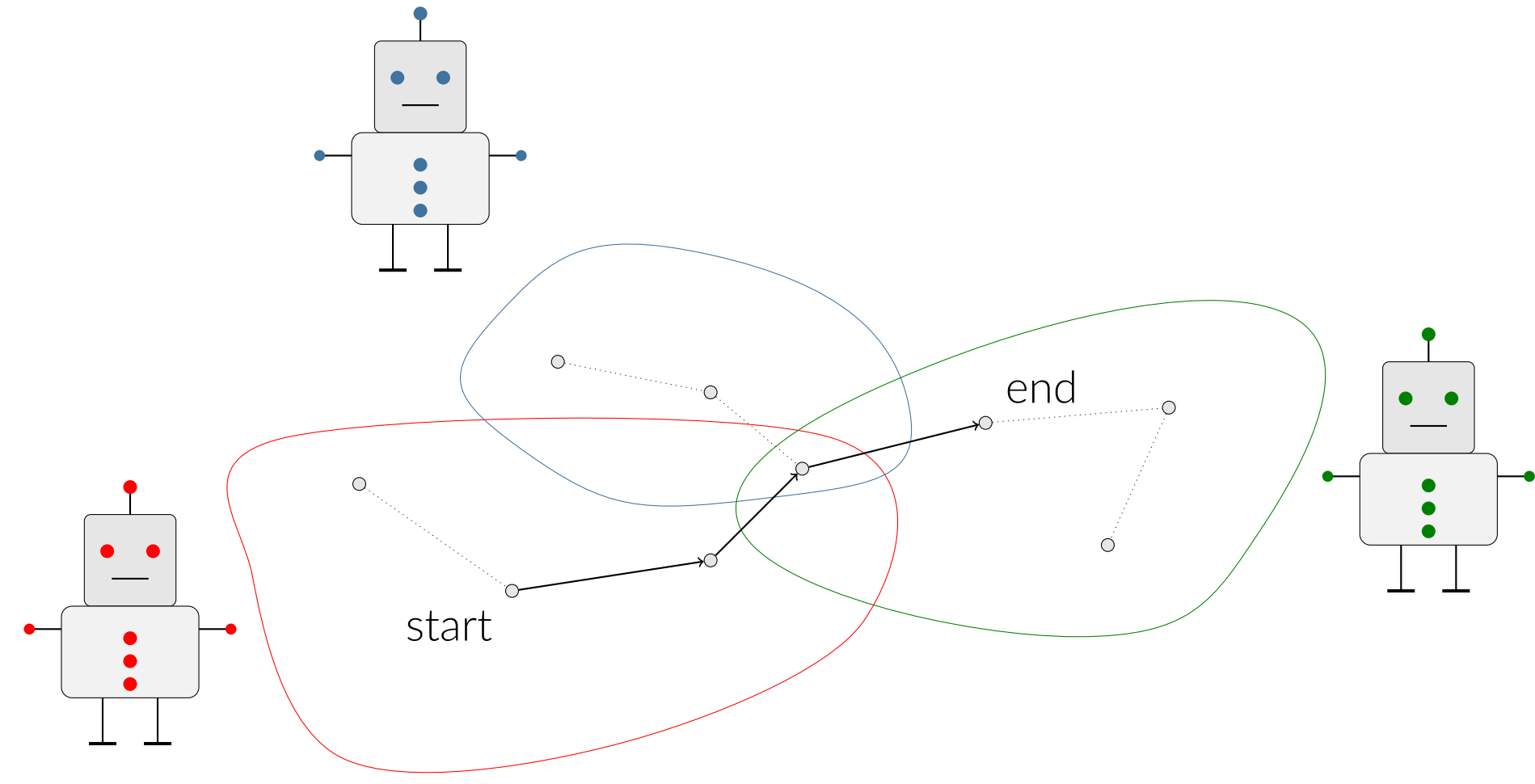


Figure 1. Path in Agent Owned Graphs

Communication Complexity

Communication Complexity studies how much information two (or more) parties must exchange in order to solve a problem when the input is split between them.

The key idea is that instead of asking: *How fast is the algorithm?*

We ask: *How many bits do the parties in the algorithm need to communicate?*

Two Party Communication Complexity

There are two players, traditionally called **Alice** and **Bob**.

- Alice holds input x
- Bob holds input y

Neither of them knows the inputs of the other persons, so they must talk back and forth until they know enough to compute some function $f(x, y)$

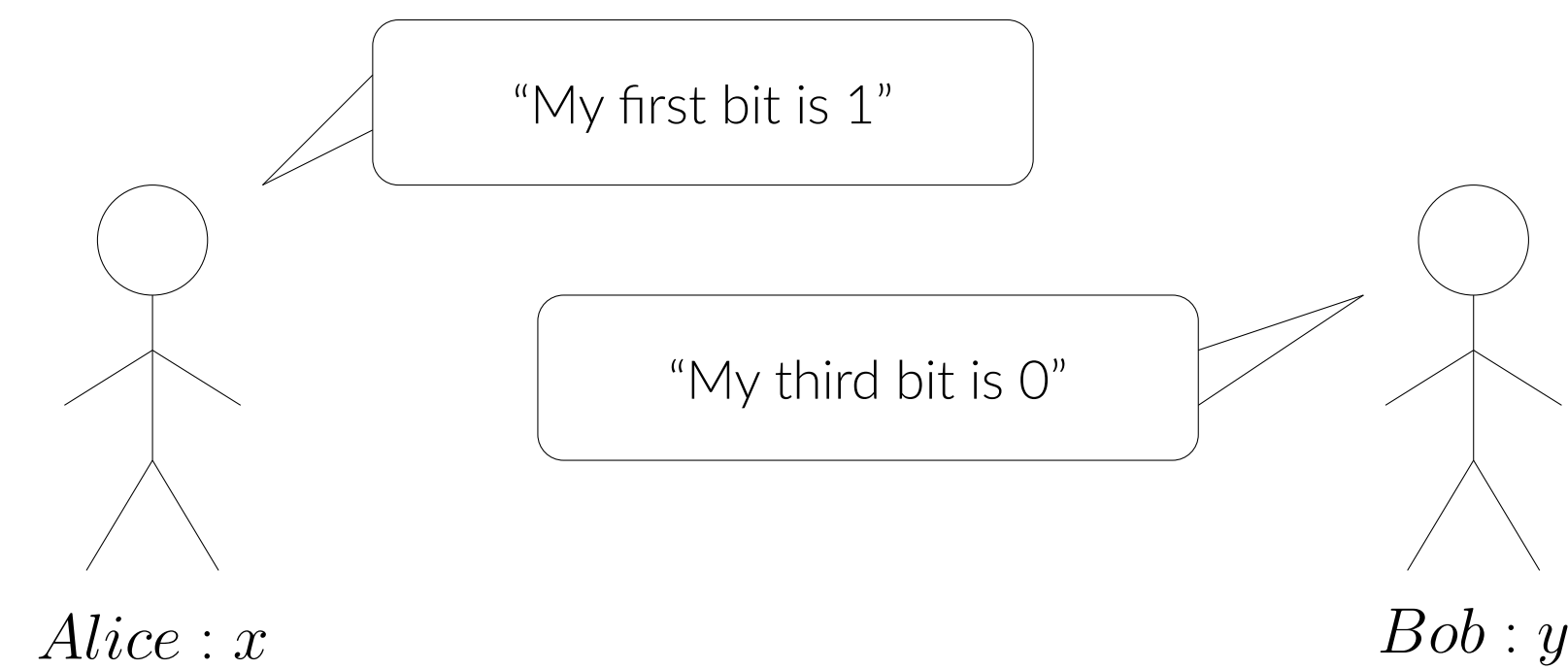


Figure 2. Alice and Bob exchanging messages.

In the worst case, someone just sends everything (n bits). So any problem that still requires n bits is **maximally hard**.

Equality is Maximally Hard

Equality Problem: Do Alice and Bob have the same n -bit string?

$$\text{EQ}(x, y) = \begin{cases} 1 & x = y \\ 0 & x \neq y \end{cases}$$

In figure 3, we represent EQ as a communication matrix:

- Rows = Alice's input x
- Columns = Bob's input y
- Blue dots mark where $\text{EQ}(x, y) = 1$

In communication complexity, a correct protocol must divide this matrix into monochromatic rectangles that contain either all 0's or all 1's.

But each blue dot is isolated, so in order to cover all 2^n diagonal 1's, the protocol needs at least 2^n rectangles. It turns out this implies that Equality requires at least n bits of communication.

Thus, **Equality is maximally hard**.

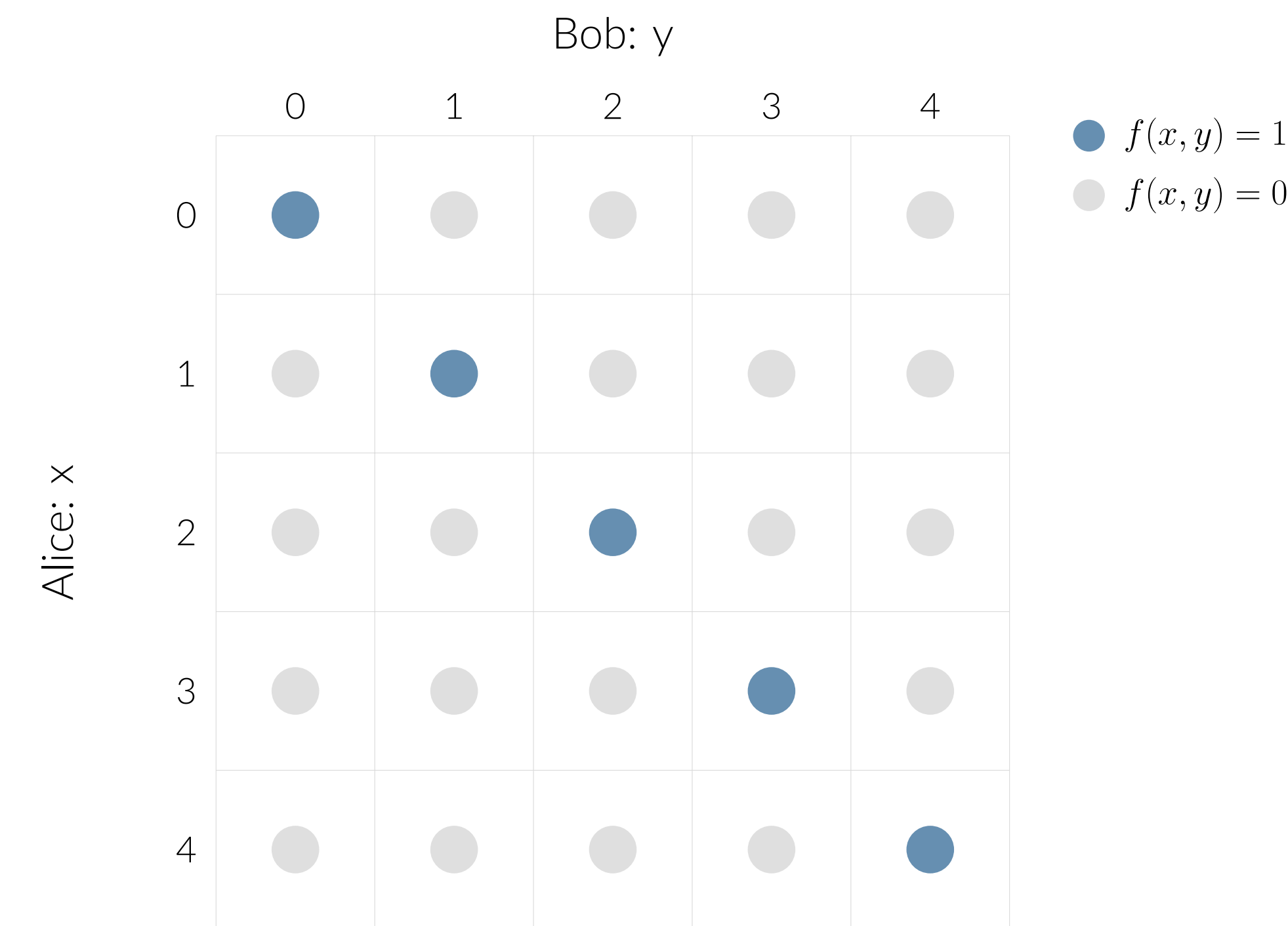


Figure 3. Communication matrix for Equality

The TRAV Problem

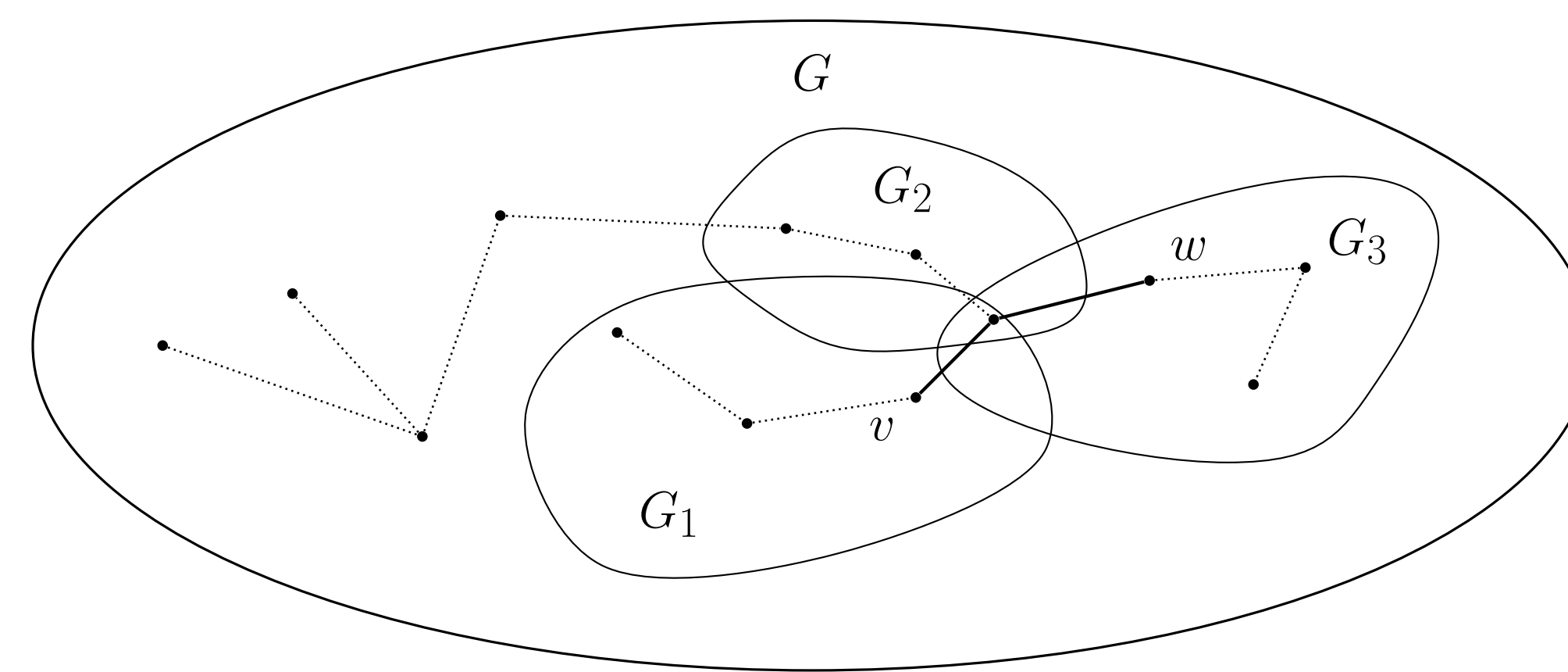


Figure 4. TRAV example where a valid path from v to w is shown in bold.

In this section, we define a simpler version of the problem from our motivation.

Consider a large graph G . Inside G , each agent "owns" part of the graph. We call these subgraphs $G_1, G_2, \dots, G_k$. No single player sees the entire graph.

Given a starting vertex v and a target vertex w , how much information do the agents need to share about their subgraphs to determine whether there is a path from v to w ?

For this poster, we will only consider a two agent case.

The Equality Gadget

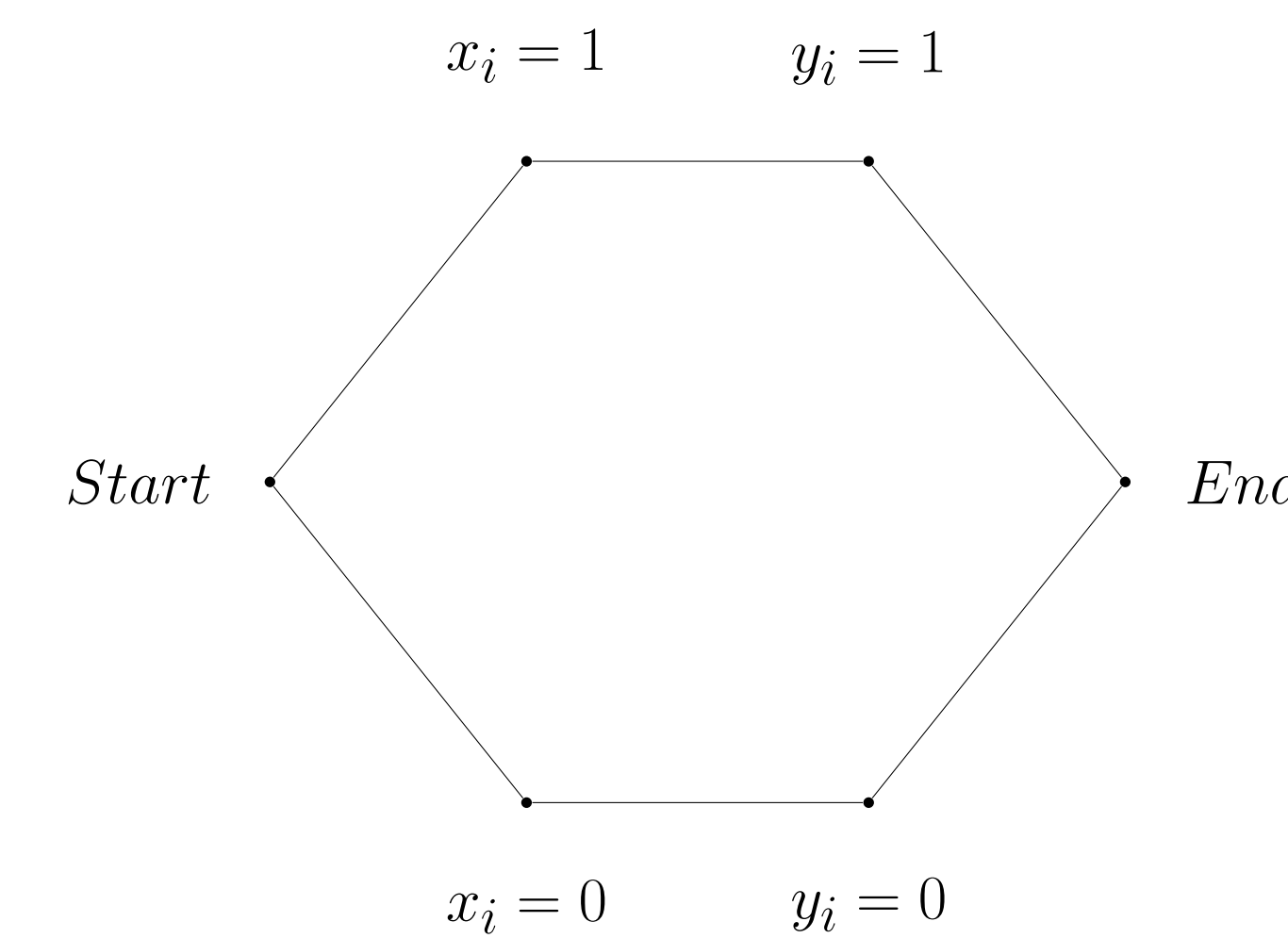


Figure 5. Gadget for proving equality Embedding

1. Top branch corresponds to $x_i = y_i = 1$,
2. Bot branch corresponds to $x_i = y_i = 0$.

Consider this gadget where its only traversable from Start to End if both x_i and y_i are both 0 or 1. In other words when $y_i = x_i$

Embedding Equality into TRAV: Equality Gadget Chain

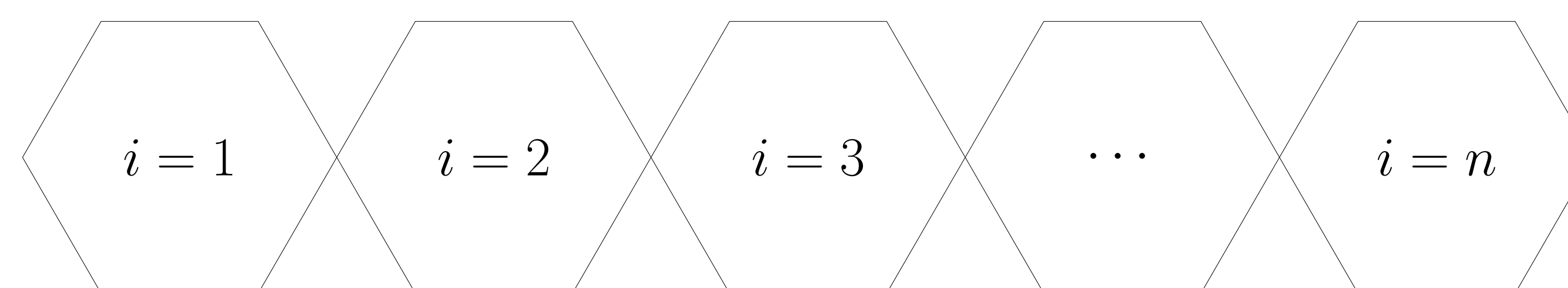


Figure 6. Chain of Equality Gadgets

To prove that TRAV is also maximally hard, like Equality, we show that we can reduce Equality to TRAV.

By chaining the gadgets together, we turn each bit comparison into a small "gate." The chain is traversable only if every gate opens, which happens exactly when $x_i = y_i$ for all i , in other words, when $x = y$.

Therefore, any protocol that solves TRAV can also solve Equality. Since Equality is deterministically maximally hard, **TRAV is also deterministically maximally hard**.

Introducing Randomness

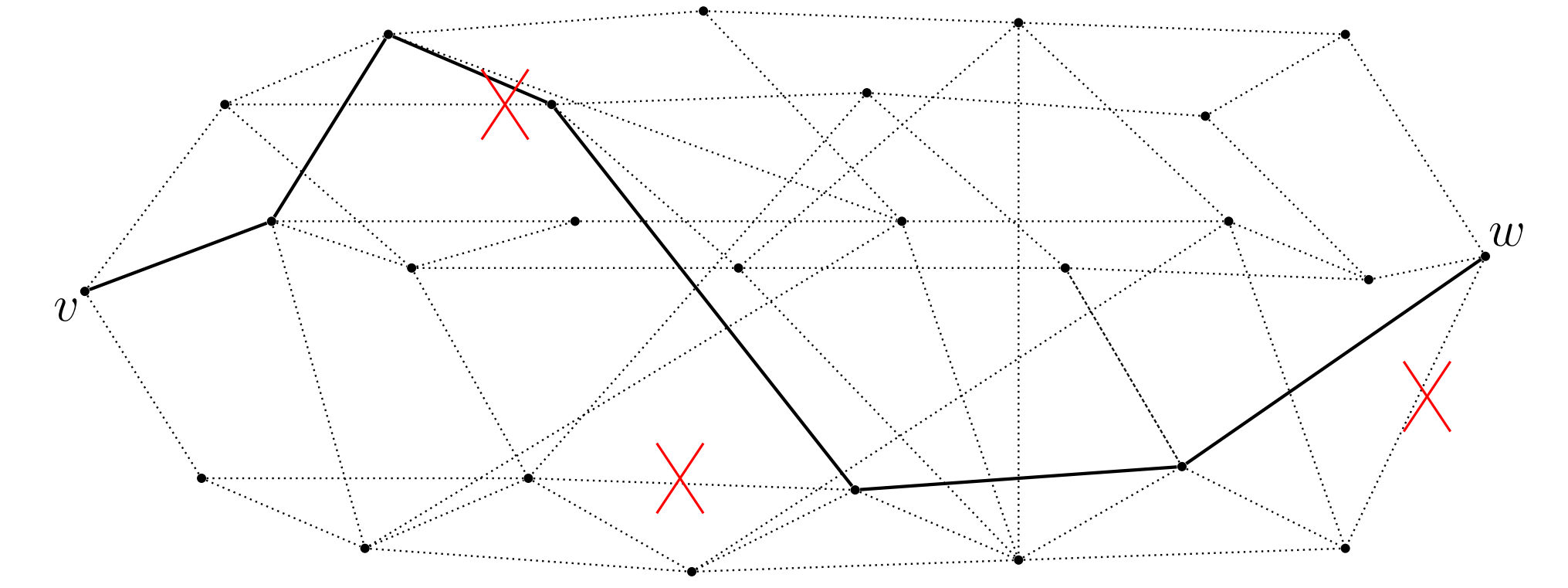


Figure 7. Graph with Skipped Edges

Let's think about another approach. What if we allow randomness and small errors? Could the agents guess parts of each other's edges and ignore some information but still get the right answer most of the time? Maybe randomness lets us communicate much less.

Disjointness is Still Hard — Even with Randomness

Unfortunately, for Disjointness, randomness does not save us.

Think of the problem like this: Alice and Bob each hold a set of items. They want to know whether they share an item.

If they do share even one item, the answer is NO (not disjoint). If they share none, the answer is YES (disjoint).

The difficulty is that the "witness" — the shared element — could be in any one of the n positions.

Random guessing does not help much, because missing that one position leads to the wrong answer.

In fact, it has been proven that even allowing randomness and a small probability of error, Disjointness still requires $\Omega(n)$ communication.

Disjoint Gadget and Embedding

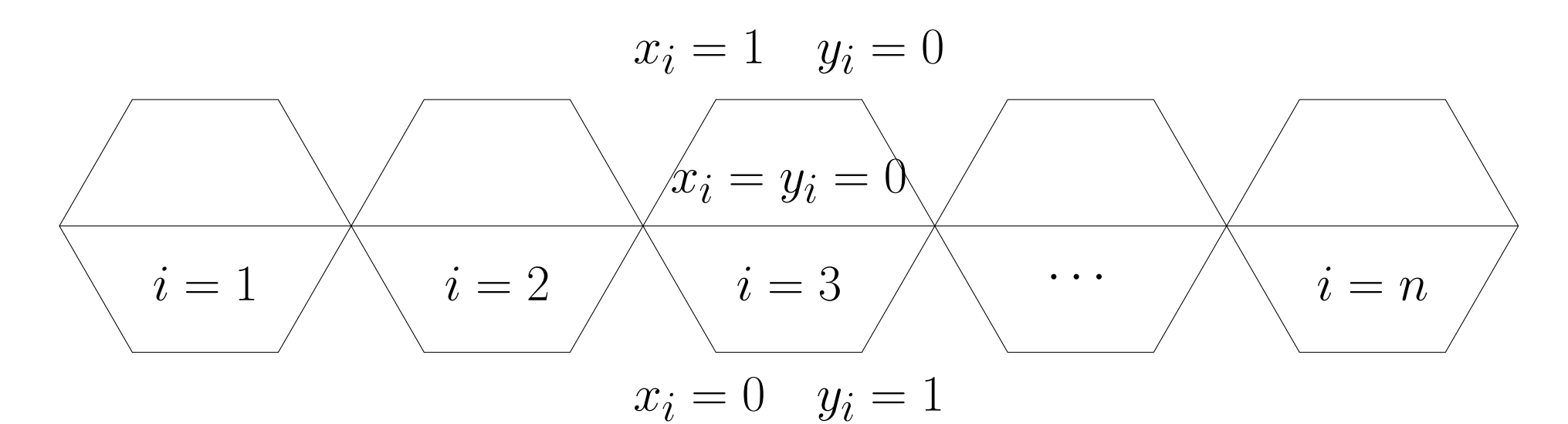


Figure 8. Chain of Disjoint Gadgets

1. Top branch corresponds to x_i having the element but not y_i .
2. Middle branch corresponds to both x_i and y_i not having the element.
3. Bottom branch corresponds to y_i having the element but not x_i .

The logic of the disjoint gadget chain is similar to the equality construction. When we chain the gadgets together, there exists a path from the first gadget to the last if and only if every gadget is traversable, that is, if and only if $x \cap y = \emptyset$